

Sanjuksha Nirgude

Autonomous Drone Systems · Speaker · Researcher · Judge · IEEE RAS · Women in Robotics Organizer

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INTRODUCTION

Robotics and automation leader with close to a decade of experience across **aerial, mobile, and multi-robot systems**, spanning simulation, autonomy, and real-world deployments. Currently leads development of advanced autonomous drone systems at Nightingale Security, enabling mission-critical operations for law enforcement, government agencies, and private security organizations. Research contributions span swarm intelligence, robot perception, and motion planning — with **5 arXiv preprints and 9+ citations**. Beyond technical practice, actively shapes the broader robotics ecosystem through **speaking as a guest lecturer at universities and podcasts, mentorship and leadership** within the **Women in Robotics community**, and as a **judge** for global robotics competitions.

TECHNICAL SKILLS

Languages: C++, Python

Software & Tools: ROS/ROS2, Linux, Git, Gazebo, Docker, Rviz, ATAK, MoveIt, Doxygen, LaTeX, AWS , ATAK

EDUCATION

Worcester Polytechnic Institute (WPI) <i>Master of Science in Robotics Engineering</i>	GPA: 4.0/4.0	Worcester, MA May 2019
University of Pune (UoP) <i>Bachelor of Mechanical Engineering</i>	Percentage: 71/100	Pune, India June 2016

WORK EXPERIENCE

Nightingale Security *Autonomy Technical Lead* 2024 – Present

- Technical owner for **ATAK integration** with Nightingale Drones, implementing **multi-drone control and monitoring from a single client**; deployed and supported defense and major law enforcement customers, maintaining compatibility across ATAK versions.
- Architected and deployed production-grade **UAV simulation infrastructure** for large-scale testing and system validation, enabling safe development of mission-critical systems without reliance on physical hardware.
- Developing **multi-drone aerial robotics systems** designed for continuous security operations, with emphasis on system reliability, robustness, and operational safety.
- Led **end-to-end drone provisioning, automation, and deployment readiness**, including testing, configuration, and documentation, eliminating single-point dependencies and standardizing operational workflows.
- Designed and implemented a structured **customer support and on-call model** across defense, law enforcement, and energy verticals, improving response efficiency and enabling the team to develop independent expertise across all accounts.
- Enabled team scalability through **technical mentoring, documentation, and knowledge transfer**, serving as escalation point for complex system-level issues; contributed to hiring through resume screening and technical evaluations.

ClearPath Inc *Senior Simulation Developer* 2023

- Cloud-based Gazebo simulation (AWS EC2) for warehouse robots; led pallet pick-and-place feature from design to executive demo.
- Acted as team triage lead, interfacing across engineering and stakeholder teams to scope and prioritize technical issues.

Open Robotics *Software Engineer – Robotics* 2022

- Contributed to ROS/Gazebo (Garden release) for NASA's VIPER lunar rover — part of a select group of engineers worldwide building next-generation lunar surface autonomy simulation.
- Implemented rendering bug fixes in Gazebo camera packages; conducted rigorous code review of open-source contributions to ensure mission-critical reliability.

- Led **AMR case-under-pick capability** development; contributed to pick/place algorithms using both conventional and **AI-based control strategies**, validated in a test-driven environment.
- Optimized code scalability to handle simulations with over **100 robots simultaneously**; built real-time diagnostic tooling for sensor data analysis and localization problem resolution.

Waypoint Robotics Inc *Robotics Intern*

2018

- Developed AMR behaviors using LIDAR, digital IO, and motion planning; expanded the programming environment and UI; integrated deep learning-based machine vision enabling live, Natural Interaction-type robot control from video input.

Cere Labs Pvt Ltd *Machine Learning Intern*

2016

- Applied **Q-learning** reinforcement learning to autonomously guide a crawling robot via **Raspberry Pi**; authored and published a practitioner blog series documenting the full build and algorithm implementation.

SELECTED RESEARCH

5 *arXiv preprints (cs.RO / cs.CV)* · 9+ citations · Work conducted 2017–2019 at WPI

- Nirgude, S., Nema, A., Jagetia, A. “**Occlusion-Based Cooperative Transport for Concave Objects with a Swarm of Miniature Mobile Robots.**” *arXiv:2409.02436* [cs.RO]. arxiv.org/abs/2409.02436
- Nirgude, S., DuCharme, K., Madhusoodanan, N. “**Detection, Recognition and Pose Estimation of Tabletop Objects.**” *arXiv:2409.00869* [cs.CV, cs.RO]. arxiv.org/abs/2409.00869
- Nema, A., Grontkowski, C., Calzada, D., Nirgude, S. “**Automated Cinematography Motion Planning for UAVs.**” *arXiv:2409.00864* [cs.RO]. arxiv.org/abs/2409.00864
- Kumar, A., Sahasrabudhe, A., Nirgude, S. “**Fuzzy Logic Control for Indoor Navigation of Mobile Robots.**” *arXiv:2409.02437* [cs.RO]. arxiv.org/abs/2409.02437
- Kumar, A., Sahasrabudhe, A., Perugu, C., Nirgude, S., Murugan, A. “**Kinematics & Dynamics Library for Baxter Arm.**” *arXiv:2409.00867* [cs.RO]. arxiv.org/abs/2409.00867

LEADERSHIP, OUTREACH & SPEAKING

- **UC Berkeley, Invited Speaker** — “The Future of Physical Security Is AI & Autonomy”; talk on AI in Robotics delivered to graduate and professional audiences
- **NASA L’SPACE Program, Subject Matter Expert** — Invited to provide expert evaluation and guidance in autonomous simulation systems for NASA-affiliated student teams
- **SICK USA Podcast, Guest Expert** — Thought leadership on autonomous mobile robot advancements and career pathways in robotics; national industry reach
- **Robots.net, Featured Expert** — Spotlights contributions to autonomous robotics and security drone innovation
- **Women in Robotics Bay Area, Organizer & Speaker** — Organized 15+ events across Boston and SF Bay Area chapters
- **FIRST Robotics Competition, Judge** — Evaluates student engineering teams on technical design, innovation, and systems integration at regional competitions
- **Industry Mentor, Women Coding Community** — Advises engineers on research-to-industry transition and real-world application of robotics

AWARDS & RECOGNITION

- **Galaxy of PCCOE Stars** — Featured in alumni recognition book by Pimpri Chinchwad College of Engineering, Pune, for outstanding contributions to the robotics field
- **ABU Robocon – Asia Pacific Robot Contest, 4th Place (Super League), India** — Designed and built two badminton-playing mobile robots; team ranked 4th nationally at University of Pune, competing in the Asia-Pacific Broadcasting Union’s international robotics contest

AREAS OF EXPERTISE

Autonomous Aerial Systems & UAS
 Multi-Drone Coordination & Swarm Intelligence
 Robot Perception & Computer Vision
 Motion Planning & Navigation
 Simulation & Digital Twin Development
 Physical AI & Embodied Intelligence
 Research-to-Deployment Feasibility

Real-Time Video Streaming & Sensor Fusion
 Robot Networking & Communication Protocols
 Human-Robot Interaction & Operator Interfaces
 Behavior Trees & State Machine Design
 Fuzzy Logic & Behavior Planning
 Edge Computing for Autonomous Systems
 ROS/ROS2 Infrastructure Management